# MASENO UNIVERSITY

## UNIVERSITY EXAMINATIONS 2017/2018

THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY

## MAIN CAMPUS

## SPH 303: QUANTUM MECHANICS I

Date: 26th February, 2018 Time: 8.30 - 11.30am

**INSTRUCTIONS:**

- Answer Question ONE and any other TWO in Section B.

Answer question 1 (Compulsory) and any other two questions

## SECTION A

### *QUESTION 1 (30 MARKS)*

a) (i) Write down the quantum operators for position vector $\vec{r}$, linear momentum $\vec{p}$, orbital angular momentum $\vec{L}$.

(ii) Show that the total energy of a quantum operator can be derived from the time-dependent Schroedinger equation of the form $\hat{E} = i\hbar\frac{\partial}{\partial t}$, where $t$ is the time.

(iii) Transform the total energy conservation equation for a particle of mass $m$ moving along the x-axis in a potential $V(x)$ in the form $E = \frac{p^2}{2m} + V(x)$ into the corresponding quantum mechanical equation.

b) (i) If in a Young's double-slit experiment, the probability amplitudes for light to pass through slits $S_1, S_2$ are $\psi_1(\vec{r}_1), \psi_2(\vec{r}_2)$ respectively, determine the interference term in the probability density for light to pass through either slit. Assuming the probability amplitudes, $\psi_1(\vec{r}_1)$, $\psi_2(\vec{r}_2)$ are complex, determine the maximum and minimum values of the intensity observed on the screen.

(ii) Suppose a wave function $\psi(\vec{r})$, is obtained as a superposition of normalized and orthogonaleigenfunctions $\psi_1(\vec{r}), \psi_2(\vec{r}), \psi_3(\vec{r})$ in the form $\psi(\vec{r}) = \alpha_1\psi_1(\vec{r}) + \alpha_2\psi_2(\vec{r}) + \alpha_3\psi_3(\vec{r})$, where $\alpha_1, \alpha_2, \alpha_3$ are complex numbers, determine the normalization condition for $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 = 1$

c) (i) Show that quantum operators which do not commute cannot have a common eigenfunction.

(ii) Determine the minimum value of Heisenberg's uncertainty product in the measurement of mean position coordinate $\hat{x}$ and linear momentum $\hat{p}$ given the commutation relation $[\hat{x}, \hat{p}] = i\hbar$

(iii) Determine the mass of a free particle of de Broglie wavelength $\lambda = 3\times10^{-11} m$ and kinetic energy $10eV$. Take Planck constant $\hbar = 6.5819\times10^{-22} MeV$ sec

## SECTION B

**Answer any TWO questions in this section.**

### *QUESTION 2 (20 MARKS)*

a) (i) Derive the de-Broglie wave-particle duality relation.

(ii) Show that the de-Broglie wave-particle duality relation provides the relation between linear momentum $\vec{p}$ and wave vector $\vec{k}$.

b) Consider a particle of mass $m$ in a plane wave motion described by a wave function $\psi(x) = A\sin(kx - \omega t)$, where $x$ is the displacement along the $x-$ axis, $\omega$ is the angular frequency, $k$ is the wave number and $t$ is the time.

(i) Show that $\psi(x)$ is an eigenfunction of the kinetic energy operator and obtain the kinetic energy of the particle.

(ii) By evaluating the commutation bracket $[\hat{x}, \alpha\hat{p}^3]$ where $\alpha$ is a constant parameter and $\hat{p}$ is the linear momentum, while $\hat{x}$ is the displacement operator, obtain Heisenberg's uncertainty relation in the measurements of the mean values of displacement $\hat{x}$ is the displacement operator, obtain Heisenberg's uncertainty relation in the measurements of the mean values of $\hat{x}$ and $\alpha\hat{p}^3$.

(iii) If the particle is confined to move between points $x = -2$ and $x = 2$ along the x-axis, apply boundary and normalization conditions to determine the constant amplitude $A$ of the wave function $\psi(x)$.

### *QUESTION 3 (20 MARKS)*

The Hamiltonian of a one-dimensional quantized harmonic oscillator of mass $m$, angular frequency $\omega$ and displacement $x$ is obtained in the form $H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2$, which generates the time-independent Schroedinger equation $H\psi(x) = E\psi(x)$.

(a) By applying a factorization procedure ,

(i) determine the ground state energy and ground state wave function.

(ii) obtain the general wave function for all the energy levels as a recurrence relation.

b) By applying the annihilation and creation operators $\hat{a}$, $\hat{a}^{\dagger}$ arising from the factorization on a general wave function, derive the algebraic operations in the form $\hat{a}\phi_n(x) = \sqrt{n}\phi_{n-1}(x)$;

$\hat{a}^{\dagger}\phi_n(x) = \sqrt{n+1}\phi_{n+1}(x)$

## *QUESTION 4 (20 MARKS)*

A particle of mass $m$ in one-dimensional motion along the x-axis in a constant potential $V_1$ approaches a potential barrier specified by potential energy $V_2$ where $V_2 > V_1$.

(a) (i) Obtain the general time-independent wave function describing the particle motion in the entire region specified by potential energies $V_1$ and $V_2$.

(ii) Determine the barrier penetration or transmission probability for the particle to be found in the region beyond the potential barrier.

b) Determine the form of the wave function in the region beyond the barrier if the potential if the potential $V_2$ is greater than the total energy of the particle and use the wave function to account for quantum tunneling phenomenon.

## *QUESTION 5 (20 MARKS)*

a) Consider the time-independent Schroedinger equation of the form $H\psi = E\psi$ where $H$ is the Hamiltonian, $E$ is a constant and $\psi$ is the time-independent wave function. Determine the physical meaning of $E$

b) A one-dimensional initial wave packet with mean wave number $k_x$ and Gaussian amplitude is given as $\psi(x,0) = Ce^{\left(-\frac{x^2}{4(\Delta x)^2}+ik_x x\right)}$, $C$ is a constant.

(i) Calculate the corresponding $k_x$ distribution.

(ii) Calculate the time dependent probability density $|\psi(x,t)|^2$

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THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN INDUSTRIAL CHEMISTRY WITH INFORMATION TECHNOLOGY

## MAIN CAMPUS

## SPH 304: PHYSICS OF THE ENVIRONMENT AND ENERGY RESOURCES

Date: 16th February, 2018 Time: 3.30 - 6.30 pm

**INSTRUCTIONS:**

- Answer a total of THREE questions only
- Question ONE is compulsory and must be attempted by all candidates

# SPH 304 Physics of the Environment and Energy resources

Sermester I, December 2017 examination

**Answer a total of three Questions only.**

**Question 1 is compulsory and must be attempted by all candidates.**

**Constants:**

Stefan-Boltzman const; $\sigma = 5.67 \times 10^{8}\ Wm^{-2}K^{-4}$

Planck constant, $h = 6.626 \times 10^{-34} J\ s$

## QUESTION ONE

(a) (i). State the reaction that produces energy from the sun. (2mks)

(ii). Explain what is meant by "solar constant" (2mks)

(iii). Give an expression for energy received from the sun per wavelength. (2mks)

(b). Other than hydro power, name two other sources of energy from water. (2mks)

(c). Briefly explain geothermal energy occurrence. (2mks)

(d). (i). Explain the importance of ozone layer in the atmosphere. (2mks)

(ii) How is ozone level maintained in the absence of a net depleting agent. (2mks)

(iii). Which atmospheric pollutant is most effective in depleting ozone layer (2mks)

(e). Describe at least two types of adiabatic lapse rates in the atmosphere and how they dictate atmospheric processes. (2mks)

(f).(i). Explain perturbation factor as applied to wind energy conversion. (2mks)

(ii). Give the theoretical minimum value of the perturbation factor. (2mks)

(g). Explain what is meant by "dynamic matching" in wind energy application. (2mks)

(h). Discuss the forces that create global wind. (2mks)

(i). Explain the occurrence of lightning (2mks)

(j). Explain one method of determining the temperature of the sun (2mks)

## QUESTION TWO

(a). What are the advantages of non-conventional energy resources and why are they not used in large scale. (5mks)

(b). Using appropriate diagram, explain how sunlight is converted directly into electricity. Why is the efficiency of this process low? (10mks)

(c). Use the following information to estimate the temperature of the sun:

sun-earth distance is $1.5 \times 10^{8}$ km

the radius of the sun is $7.5 \times 10^{5}$ km

Solar radiation reaching the earth's atmosphere is 1400 $Wm^{-2}$ (5mks)

## QUESTION THREE

(a). Describe the formation of acid rain and explain how the various elements get into the atmosphere. (6mks)

(b). Sketch on the same diagram solar radiation spectrum at AM0, AM1 and AM2 and explain the differences. (6mks)

(c) .A mass of air with a temperature of $1^{o}K$ above the temperature of saturated environment which is at $10^{o}C$ and mixing ratio of $6gkg^{-1}$ is ascending without mixing. What is the acceleration of this air? (4mks)

(d). Explain geothermal energy occurrence and how it is applied in electricity generation. (4mks)

## QUESTION FOUR

(a). Explain what is meant by coriolis force . (5mks)

(b). Energy in the wind can be used to perform mechanical work which in turn can be used to generate electricity. Define perturbation factor of wind power extraction. (5mks).

(c). Using linear momentum theory, clearly define the key parameters in wind power extraction process and show that the power extracted from wind is give n by:

$$P_T = 2\rho A_1 U_1^2 (U_o - U_1)$$

Show further that:

$$P_T = P_o\,[4a(1-a)^2]$$

(10mks)

## QUESTION FIVE

(a). Briefly discuss commercial applications of solar energy (6mks)

(b). A solar cell of area $25cm^2$ having a peak current and voltage of 5mA and 500mV respectively is exposed to $1000Wm^{-2}$ of solar radiation.

(i). Determine the efficiency of this cell. (7mks)

(ii). If the short circuit current is 60mA and open circuit voltage is 600mV, determine the fill factor. (7mks)

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THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY

## MAIN CAMPUS

## SPH 305: STRUCTURES AND PROPERTIES OF MATTER I

Date: 15th February, 2018 Time: 8.30 -11.30 am

**INSTRUCTIONS:**

- Answer question ONE and any other TWO questions in this paper.

## SPH 305: STRUCTURES AND PROPERTIES OF MATTER I

## INSTRUCTIONS

Attempt Question **One** and any other Twoquestions in this paper.

### Useful Constants

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Surface tension of water $\gamma = 6.5 \times 10^{-2} N/m$

Universal gas constant $R = 8.31 J/mol/K$

Density of water $1000 Kg/m^3$

Boltzmann constant $K = 1.38 \times 10^{-23} J/K$

Acceleration due to gravity $g = 9.8 m/s^2$

**QUESTION ONE (30 marks)**

a) **i.** State two differences between crystalline and amorphous solids (2 marks)

**ii.** Sketch the planes; [101] and [102] within the cubic unit cell (2 marks)

b) Name any two of the 3-D Bravais lattices (2 marks)

**ii.** What causes dipoles in the molecules of a substance (2 marks)

c) In Covalent and Vander Waal solids the resultant force of attraction between the constituent particles is given by: $F = \frac{A}{x^m} - \frac{B}{x^n}$. Identify the repulsive and attractive components of this force

(2 marks)

d) If first-order reflections occurs in a crystal at Bragg's angle of 3.4°. At what angle does second-order reflections occur from the same family of reflecting surfaces(3 marks)

e) A bar of length L, cross section area A and Young's modulus E is subjected to a tension F, if the stress and strain in the bar are represented by Q and P respectively. Derive an expression for the elastic potential energy per unit volume of the bar in terms of P and Q. (3 marks)

f) Although glass has higher tensile strength than bricks, it is more brittle. Explain why? (2 marks)

**QUESTION FOUR (20 marks)**

a) A certain metal has BCC crystal structure. If the angle of diffraction for the(211) set of planes occurs at 75.99° (1st order reflection when monochromatic X-ray radiation of wavelength 16.59 nm is used. Calculate

i. the interatomic spacing for the set of planes (4 marks)

ii.the atomic radius of its atom (4 marks)

b) The attractive forces between atoms are basically electrostatic in origin and the classification of different types of bonding is strongly dependent on the electrostatic structure of the atoms. Describe the four classes into which these attractive bonds can be divided giving an example of each (8 marks)

c) A pipe 2.5 cm in diameter is full of water flowing at 1.8 m/s,if the pipe narrows to 2.0 cm diameter, Calculate the flow speed at the narrow section? (Take density of water as 1000kg/m$^3$) (4 marks)

**QUESTION FIVE (20 marks)**

a)An ideal gas at 27°C and pressure 760mmHg is compressed reversibly until its volume is halved. Calculate the final pressure and temperatureofthe gas if the gas is

i in the case of isothermal compression (5 marks)

ii.foran adiabatic compression(Given the heat capacities of the gas are $C_p = 2100 J/Kg/K$ and $C_v = 1500 J/Kg/K$) (6 marks)

b) The energy per unit moleof a crystal is given by $U(r) = N_A\left[\frac{-Ae^2}{4\pi\varepsilon_o r} + \frac{B}{r^n}\right]$ show that the minimum energy of the crystal for nearest neighbor distance $r_o$ is given by $U(r_o) = \frac{-Ae^2 N_A}{4\pi\varepsilon_o r_o}\left[1 - \frac{1}{n}\right]$ where $N_A$ is the Avogadro's number. (9 marks)

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## UNIVERSITY EXAMINATIONS 2017/2018

THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

## MAIN CAMPUS

## SPH 307: INTRODUCTION TO ELECTRONICS

Date: 14th February, 2018 Time: 12.00 -3.00 pm

**INSTRUCTIONS:**

- Answer question ONE in SECTION A and any other TWO questions from SECTION B

SECTION A. This section is **COMPULSORY.**

It carries a total of **30 marks.**

1. a) (i) (i) Define an oscillator. **(2 marks)**

(ii) Draw an oscillator equivalent circuit and name the parts. **(3 marks)**

(iii) Write the equation for an oscillator circuit. **(2 marks)**

b) (i) Give two characteristics of an ideal op-amp. **(2 marks)**

(ii) Define *bandwidth* and give an example of an amplifier application where it is important. **(3 marks)**

c) (i) By indicating the flow of carriers in a diagram and explaining each step, derive the expression of conductivity ($\sigma$) of an intrinsic semiconductor in terms of mobility ($\mu$). **(8marks)**

(ii) W hat is the resistivity of an intrinsic germanium semiconductor at $300^0$K given that q = $1.6x10^{-19}$C, $n_i = 2.5x10^{13}$, $\mu_n = 3600$ and $\mu_p = 1700$. **(3 marks)**

d) (i) Why is NPN transistor is more popular than PNP transistor. **(2 marks)**

(ii) In practice, why are transistors most often used in *common-emitter* configuration. **(2 marks)**

(iii) A transistor is connected in a common-emitter mode in a circuit and has collector load of 2kΩ. The short-circuit current gain of the transistor is 100 and its input resistance is 1kΩ. Calculate the voltage gain $A_v$ and power gain $A_p$ of the transistor. **(3 marks)**

**SECTION B.** Answer **ONLY TWO** questions from this section.

Each question carries twenty (20) marks.

2. a) (i) Semiconductor diodes have some important parameters. Name any three. **(3 marks)**

(ii) Explain the *avalanche effect* and *Zener effect* in a diode. **(5 marks)**

(iii) Figure 1 shows the characteristics of a Zener diode.

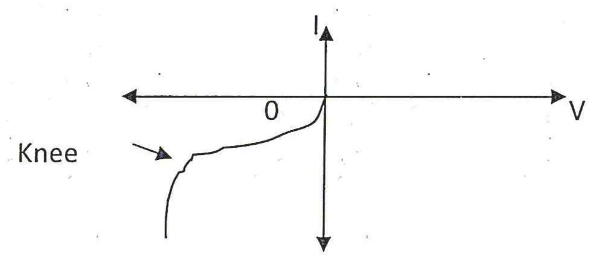


**Fig. 1**

Explain the significance of the *knee*? **(2 marks)**

b) (i) PN junctions are often used as photo detectors. Assume you have a PN junction that is reverse biased as shown in figure 2.

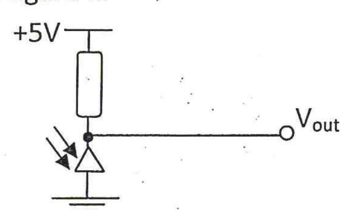


**Fig. 2**

The junction is specifically fabricated to expose the P-N interface. When light is shown is shown on the interface, the voltage $V_{out}$ changes. Describe, briefly the mechanism that causes this voltage change. **(4 marks)**

(ii) Assuming that this PN junction is made of silicon, will the voltage change occur for all frequencies of light. Explain. **(3 marks)**

(iii) If you wanted the voltage swing at the output bigger for the same light intensity, are there any changes you could make to the doping concentration of the P and N regions of the PN junction. Explain. **(3 marks)**

3. a) (i) Define a transistor. **(2 mark)**

(ii) Cite two important features of a transistor. **(2 marks)**

b) The diagram in figure 3 shows a simplified transistor circuit with a few discrete components.

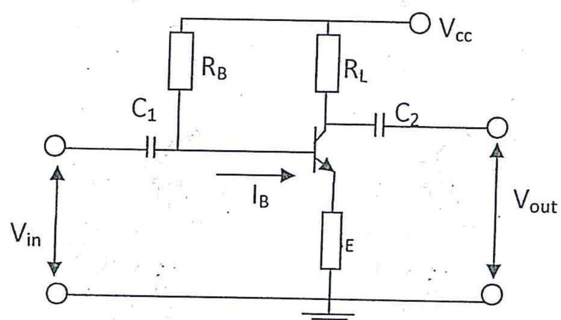


**Fig. 3**

You are given that $R_B = 1.8\ M\Omega$, $R_L = 4.7\ k\Omega$, $R_E = 1\ k\Omega$, $C_1 = C_2 = 10\mu F$ and $V_{CC} = 15V$.

(i) How are voltage $V_{out}$ and current $I_B$ are developed. **[2 marks]**

(ii) Why are capacitors $C_1$ and $C_2$ included in the circuit? **[2 marks]**

(iii) Write the *load line equation* for the circuit in figure 2. **[2 marks]**

(iv) If $V_{CE} = 7.5V$ and $\alpha_E = 50$, calculate $I_C$ and $I_B$. **[4 marks]**

c) i) In a PNP bipolar transistor, what happens if the base region was made extra wide ( i.e much wider than the diffusion length of holes). **(2 marks)**

(ii) Describe how such a device would operate and comment on the current flow. **(4 marks)**

4. a) (i) Indium antimonide has a narrower forbidden gap than silicon. But why is silicon more desirable for semiconductor devices than it? **[2 marks]**

(ii) Name four areas of application for semiconductor devices. **[2 marks]**

(iii) Semiconductor devices have many important advantages over other types of electronic devices. Name any four advantages. **[2 marks]**

b) (i) Why would electrons in the valence band of a semiconductor conduct current at room temperature. **[2 marks]**

(ii) What is a *compensated crystal*? **[2 marks]**

c) (i) What is a semiconductor? **(2 marks)**

(ii) What happens to the silicon crystal lattice if its temperature is raised above absolute temperature? **(2 marks)**

(iii) What happens to the conductivity of silicon:

I) when you dope the material with equal numbers of donors and acceptors? **(2 marks)**

II) when you dope silicon with unequal numbers of both kinds of dopants? **(2 marks)**

d) (i) The operation of a PN junction depends on controlled imperfections in a crystal. What is the purpose of the imperfections? **(2 marks)**

(ii) For conduction to be possible in a material, what conditions must be fulfilled? **(2 marks)**

5. a) Draw a schematic diagram of an operational amplifier. **[3 marks]**

b) (i) By drawing a circuit of an op-amp as an *integrator,* and showing the directions of the currents, prove that the output voltage $V_0$ and the input voltage $V_s$ of the circuit are related as:

$$V_0 = -(1/RC)\int V_s dt \qquad \textbf{[12 marks]}$$

(ii) Compute the value of $V_0$ when $V_s = V$, where V is a constant. **(2 marks)**

(iii) Where can such a voltage in (iii) above be used and why? **(3 marks)**

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**END**

# MASENO UNIVERSITY

## UNIVERSITY EXAMINATIONS 2017/2018

THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN INDUSTRIAL CHEMISTRY WITH INFORMATION TECHNOLOGY

## MAIN CAMPUS

## SPH 313: CLASSICAL MECHANICS

Date: 12th February, 2018 Time: 3.30 - 6.30 pm

**INSTRUCTIONS:**

- Answer ALL questions in SECTION A and any TWO from SECTION B.

## SPH 313: CLASSICAL MECHANICS

**Instructions**

**Section A is Compulsory**

**Answer ANY TWO Questions from Section B**

**Section A**

**This Section is Compulsory**

**Question One [30]**

a) Consider a many-particle system. Let $\vec{R}$ be the radius vector from an arbitrary point $O$ to the center of mass and let $\vec{r}_i'$ be the radius vector from the center of mass to the $i$th particle. If $\vec{r}$ is the radius vector from $O$ to the $i$th particle, show that the total angular momentum about $O$ is the angular momentum of the system concentrated at the center of mass plus the angular momentum of motion about the center of mass. **[8 mks]**

b) Distinguish between the following types of constraints:

i) Holonomic and nonholonomic **[2 mks]**

iii) Rheonomous and schleronomous **[2 mks]**

c) (i) State D'Alembert's principle. **[2 mks]**

ii) Show that D'Alembert's principle can be written as:

$$\sum_j \left\{\left[\frac{d}{dt}\left(\frac{dT}{d\dot{q}_j}\right)-\frac{dT}{dq_j}\right]-Q_j\right\}\delta q_j = 0,$$

where $q_j$ are generalized co-ordinates, $\dot{q}_j = dq/dt$ and $Q_i$ are the generalized forces.

**[10 mks]**

iii) Show that for a conservative system, the equation in c(i) above leads to Lagrange's equation. **[6 mks]**

## Section B

### Question Two [20]

We are interested in finding the extremal values of the integral

$$I = \int_{x_1}^{x_2} f(\dot{y}, y', x)dx,$$

where $f(y, \dot{y}, x)$ is a function defined on the curve $y = y(x)$ and $y' = \frac{dy}{dx}$. Show that the function $f(y, \dot{y}, x)$ that gives $I$ its extremal value satisfies the equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0.$$ **[20 mks]**

### Question Three [20 mks]

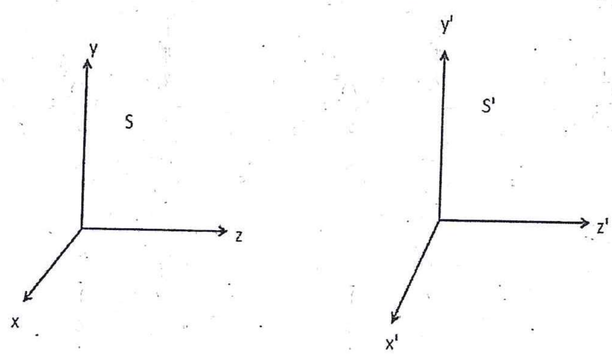
a) Using variational calculus, show that the shortest distance between two points on a plane is given by the equation of a straight line $y = ax + c$. **[8 mks]**

b) Consider a curve between two fixed points $(x_1, x_1)$ and $(x_2, x_2)$. Find the curve $y(x)$ which gives the minimal area of the surface of revolution. **[12 mks]**

### Question Four [20]

a) i) State the two postulates of Einstein's theory of special relativity. **[4 mks]**

ii) Two inertial frames of reference S and S' are shown below:




S' is moving with a velocity v relative to S along the z-axis. Assume that their origins coincide at t=t'=0 where t is the time in S and t' is the time in S'.

i) Define the Lorenz transformation. **[4 mks]**

ii) Assume that there is a source in the origin of S' that emits a light wave at t=0. Show that the equation for the spherical wave front is the same in both frames. **[6 mks]**

b) Show that the total relativistic energy, E, is given by

$$E^2=p^2c^2+m^2c^4,$$

Where p is the pomentum, m is the relativistic mass and c is the speed of light. **[6mks]**

**Question Five [20]**

a) Consider a particle moving in free space under the influence of a force. Using Lagrangian mechanics, show that the particle obeys Newton's equation of motion. **[10 mks]**

b) Consider a simple pendulum of length $\ell$ and mass $m$ moving in the x-y plane under the influence of a gravitational field as shown in the figure below. Using Lagrangian mechanics, find its equation of motion. **[10 mks]**